

CSE 232 : Data Structures

LAB 09 : MAP

Due Saturday at 5:00 p.m. MST

The ninth programming assignment of this semester is to write a class with the same functionality as the standard template library (STL) `std::map` class.

Map

Your map class will be in `map.h` and be named `custom::map`. It will need to support the following operations:

- **Construct:** default constructor, copy constructor, move constructor, initializer list constructor, and range constructor.
- **Assign:** copy assign, move assign, and initializer list assign.
- **Iterator:** begin and end (for the 100 and 115 options only).
- **Access:** subscript operator (read-only and read/write), at method (read-only and read/write), and find (100 and 115 options only).
- **Insert:** copy insert, move insert, initializer list insert, and range insert.
- **Remove:** erase element, erase iterator (100 and 115 options only), erase range (100 and 115 options only), and clear.
- **Status:** size and empty.

Three Options

This week, there will be three options.

85 Option : No Iterator

There is no iterator with the 85 option. Also, the key is always an integer and the value is always an integer.

```
BST<pair<int, int>> bst;
```

To complete the 85 option, please start with the following files: [232.09.Lab.85.zip](#).

100 Option : Iterator

This option will allow the client to specify the data-type of the key and the value.

```
template <class K, class V>
```

The underlying data structure will be the 100 option of the BST class from Lab 07:

```
BST<pair<K, V>> bst;
```

Additionally, an iterator is to be created:

- **Construct:** default constructor, copy constructor, non-default construct (so `begin()` and `end()` can properly initialize it).
- **Assign:** copy assign.
- **Compare:** equal-to and not-equal-to.
- **Access:** dereference (read-only).
- **Arithmetic:** increment (prefix and postfix), decrement (also prefix and postfix).

To complete the 100 option, please start with the following files: [232.09.Lab.100.zip](#).

115 Option : Allocations

This option will require the class to everything from the 100 option (including the iterator). This includes the same template declaration:

```
template <class K, class V>
```

Additionally, it will incorporate the BST using the Red/-Black algorithm. This version must get the correct number of allocations and frees using `spy`. Use the BST code from Lab 07 to accomplish this:

```
BST<pair<K, V>> bst;
```

To complete the 115 option, please start with the following files: [232.09.Lab.115.zip](#).

As with last week, the 115 option requires a precise number of allocations and deletions. In order to have more control over this, it is necessary to use an allocator. Please see Appendix I: Allocator in the text book for a detailed description of how this works. There is also a video demonstration here: [Allocators and Spys](#). This explains how the `spy` class works to count allocations, and how `std::allocator` can be use to more precisely control them.

Assignment

Please do the following:

1. Select an option: 85, 100, or 115.
2. Download the code.
3. Implement the data structure with your partner(s).
4. Report how long it took for each individual, and what was the hardest part.
5. Upload your `map.h` code.

Assessment

The grade for your assignment will be based on the following criteria:

	Exceptional 100%	Good 90%	Acceptable 70%	Developing 50%	Missing 0%
Code Quality 35%	The most elegant solution was found for all the methods; there is no room for improvement,	There are no errors in efficiency, maintainability, cohesion, robustness, or redundancy.	There exists one error in efficiency, maintainability, cohesion, robustness, or redundancy.	The code quality is not what is expected for a CSE 232 student.	The code is sloppy or otherwise unprofessional.

Functionality 50%	Passes all the tests	Passes 90% of unit tests	Passes 70% of unit tests	Passes 50% of unit tests	Program fails to compile or was not submitted.
Iterator 15%	The 100 or the 115 solution was attempted				The 85 solution was attempted
Allocations 15%	115 solution was attempted				The 85 or the 100 solution was attempted